



Day 2: AWS Global Infrastructure & Cloud Economics

ITNW 1009 — Where does the cloud actually live, and how do we justify the cost?

DAY 2

AWS FUNDAMENTALS

Today's Game Plan

Welcome Back — Here's What We're Building Today

By 3:30 PM, you'll be able to map global infrastructure, evaluate cloud costs, and hand a professional estimate to your manager. Let's get to work.

1

Map Global Infrastructure

Differentiate Regions,
Availability Zones, and Edge
Locations

2

Unpack Cloud Economics

Evaluate TCO savings when
moving on-premises ops to AWS

3

Build a Live Cost Estimate

Generate a multi-tier server
estimate using the AWS Pricing
Calculator

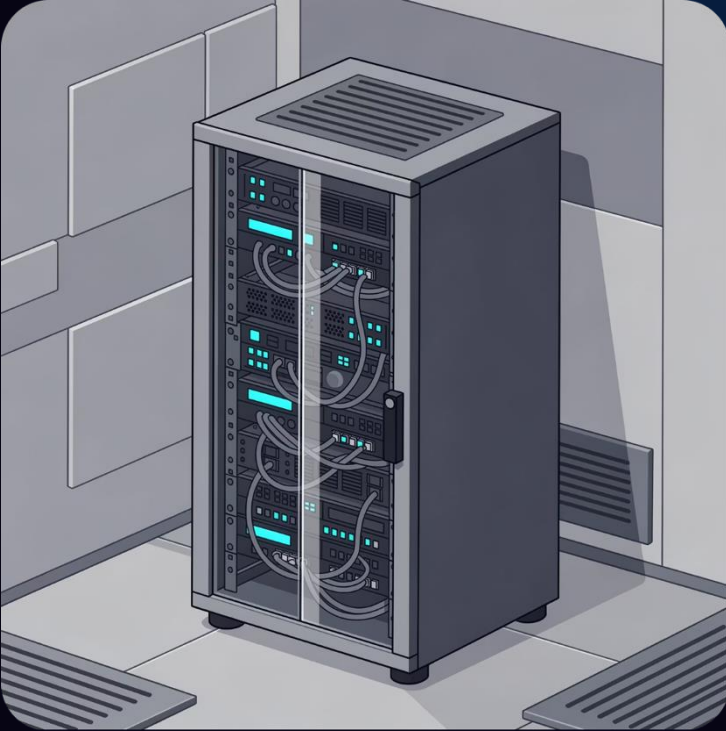


The Cloud Is Not Magic — It's Real Buildings

Every application you touch runs on physical hardware inside highly secured facilities around the world. As IT professionals, knowing **where** that hardware lives is a core career skill.

Physical Reality vs. Cloud Abstraction

From One Building to a Global Network



Traditional On-Premises

One server room. One failure point. If the building loses power — everything goes down.

AWS Global Footprint

AWS structures hardware across three foundational layers that work together to keep your applications fast, resilient, and compliant.

Regions

Availability Zones

Edge Locations

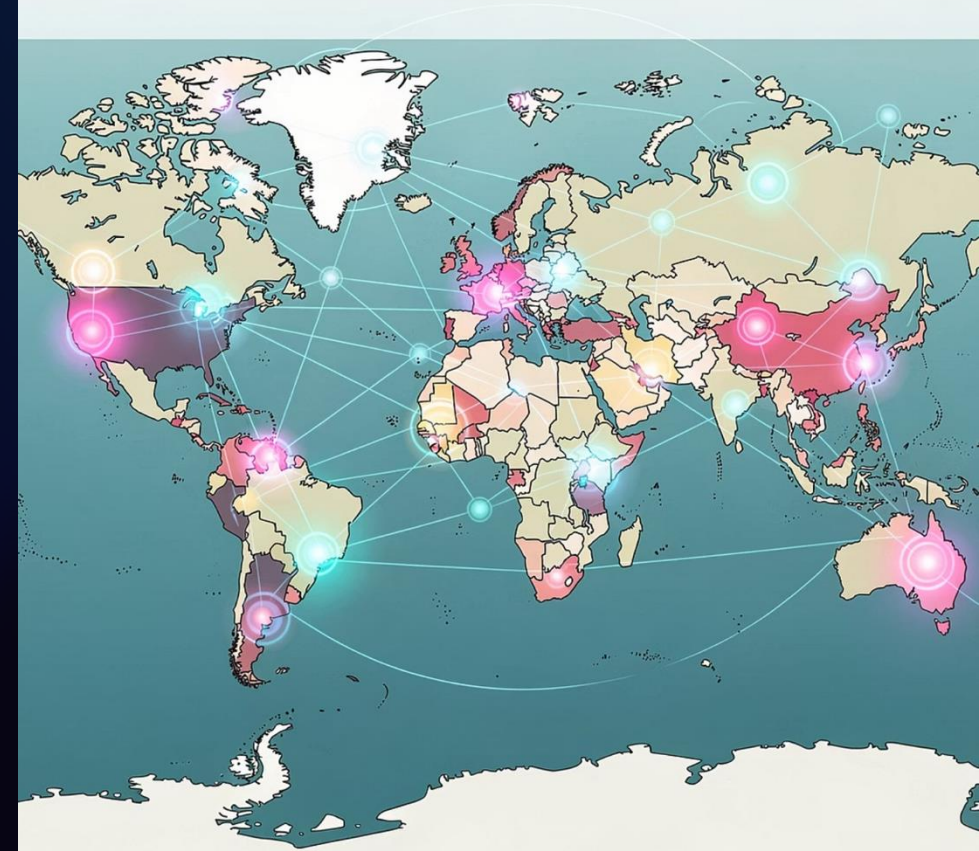
SECTION 1

Global Infrastructure

Regions · Availability Zones · Edge Locations

AWS Regions — The Big Picture

A Region is a fully isolated, distinct geographic area. AWS currently operates dozens of Regions worldwide — each one a completely independent environment with its own power, networking, and data boundaries.



Key Facts Every Technician Must Know



- **Geographically Isolated**
Each Region is completely independent. A failure in one will never cascade to another.
- **You Control Your Data**
AWS never replicates data between Regions automatically. Your data stays where you put it.
- **Named by Location & Code**
us-east-1 = Northern Virginia. **us-west-2** = Oregon. Easy to read once you know the pattern.

Key Vocabulary

Data Sovereignty — Why Region Selection Is a Legal Decision

What It Means

Data Sovereignty is the legal principle that data is subject to the laws of the country where it is stored.

Hospitals, banks, and government contractors often have legal mandates requiring customer data to stay within national borders.

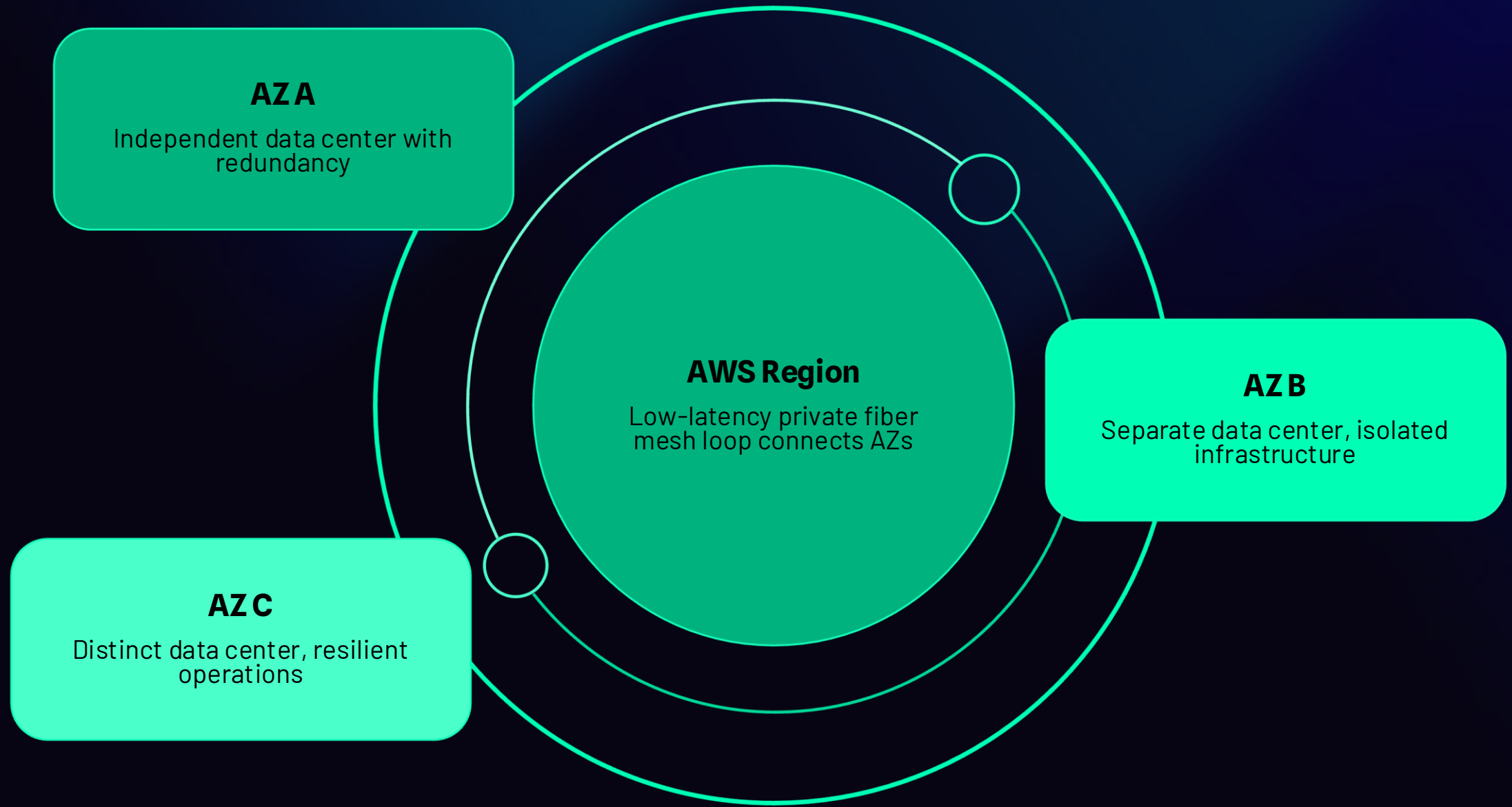
How You Solve It

Simply select the correct AWS Region when deploying your infrastructure. Choosing **eu-central-1 (Frankfurt)** for German customers keeps their data inside the EU — satisfying GDPR compliance automatically.

- ❏ Region selection is one of the first questions on any real-world cloud project. Knowing this makes you valuable on day one.

Inside a Region

Availability Zones — Built-In Redundancy



Availability Zones

Why Multiple AZs = High Availability



Imagine a flash flood knocks out the power grid serving **AZ A**. Because you deployed copies of your application in **AZ B** and **AZ C**, your company's website stays completely online.

High Availability

System remains operational despite partial failure

Fault Tolerance

Traffic reroutes automatically to healthy AZs

Quick Check

Regions vs. AZs — What's the Difference?

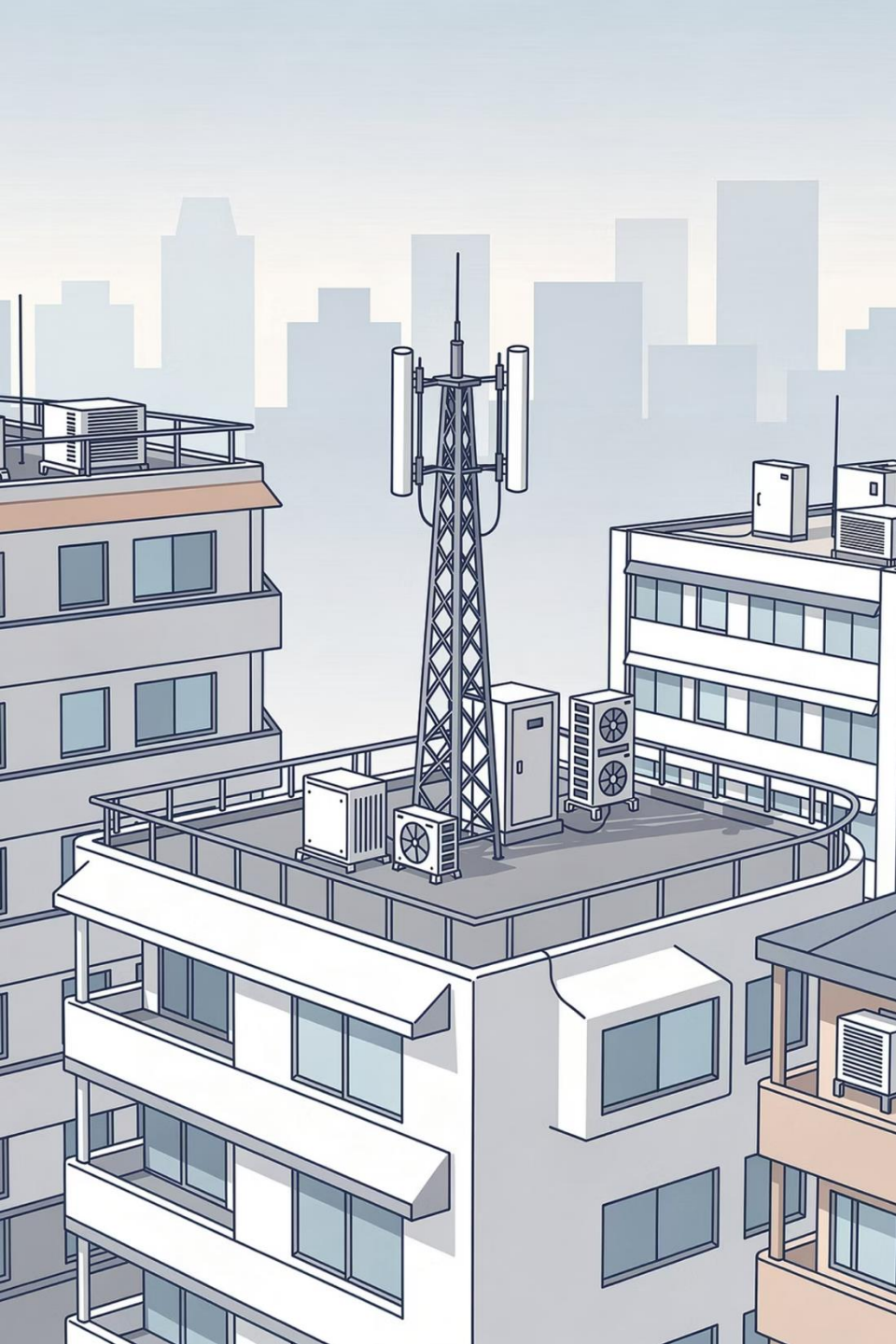
Region

A distinct geographic area (e.g., N. Virginia, Ireland). Completely isolated from other Regions. You choose this to satisfy latency and data sovereignty requirements.



Availability Zone

One or more data centers *inside* a Region, separated by distance. You deploy across multiple AZs to survive local hardware or utility failures.



Edge Locations & CloudFront

Bringing Content Closer to the User

Even the fastest cloud Region can't beat physics. When a user in London loads a video from a server in Northern Virginia, the signal travels thousands of miles of undersea fiber cable — and every mile adds delay.

Edge Locations

AWS's Network of Local Digital Refrigerators



Edge Locations cache copies of your content (videos, images, web files) at hundreds of local Points of Presence worldwide — powered by **Amazon CloudFront**, AWS's Content Delivery Network.

Edge Locations vs. AZs

Two Very Different Jobs

Availability Zone

Runs full compute, databases, and application logic.
Built for **resilience and uptime** inside a Region.

Edge Location

Caches static content (videos, images, pages) close to end users. Built for **low-latency delivery** globally.
Does NOT run application servers or databases.

📌 Think of Edge Locations like a neighborhood convenience store — stocked with what customers need nearby, not a full warehouse.

Vocabulary Check

Key Terms — Lock These In

Region

A fully isolated geographic area containing multiple AZs

Availability Zone (AZ)

Independent data center(s) within a Region, isolated from failure

Edge Location / Point of Presence

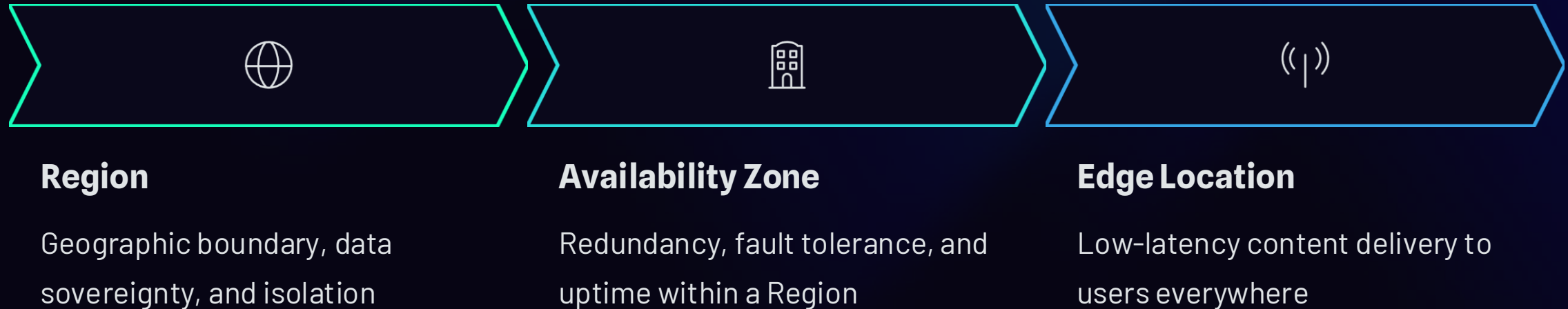
A local cache node delivering content via Amazon CloudFront

Latency

The delay (in milliseconds) caused by physical distance data must travel

Infrastructure Summary

The Three Layers Working Together



SECTION 2

Cloud Economics

CapEx · OpEx · Total Cost of Ownership



Speaking the CFO's Language

When you enter an IT department, you'll sit in meetings with financial officers who don't care about fiber optic cables. They care about **money**. Understanding cloud economics makes you the person who bridges both worlds.

CapEx vs. OpEx — The Fundamental Shift

Capital Expenditures (CapEx)

Spending money **upfront** on physical assets before you ever flip a switch.

- Server rack hardware purchases
- Building and real estate leases
- Long-term hardware amortization
- Power supply and cooling infrastructure

Operating Expenditures (OpEx)

Paying only for what you **use, when you use it** — like a utility bill.

- Monthly pay-as-you-go resource billing
- Zero upfront infrastructure commitment
- Scale up or down instantly
- Stop paying the moment you turn it off

📌 The cloud eliminates CapEx entirely and replaces it with OpEx. This is one of the most powerful financial arguments for cloud adoption.

The OpEx Mental Model

Think of AWS Like Your Electric Bill



You don't purchase a power plant from the city before turning on your bedroom light. You simply flip the switch and pay for the exact kilowatts used at the end of the month.

AWS works the exact same way. Launch a server — you pay. **Shut it down — you stop paying instantly.** No lease. No depreciation schedule. No idle overhead.

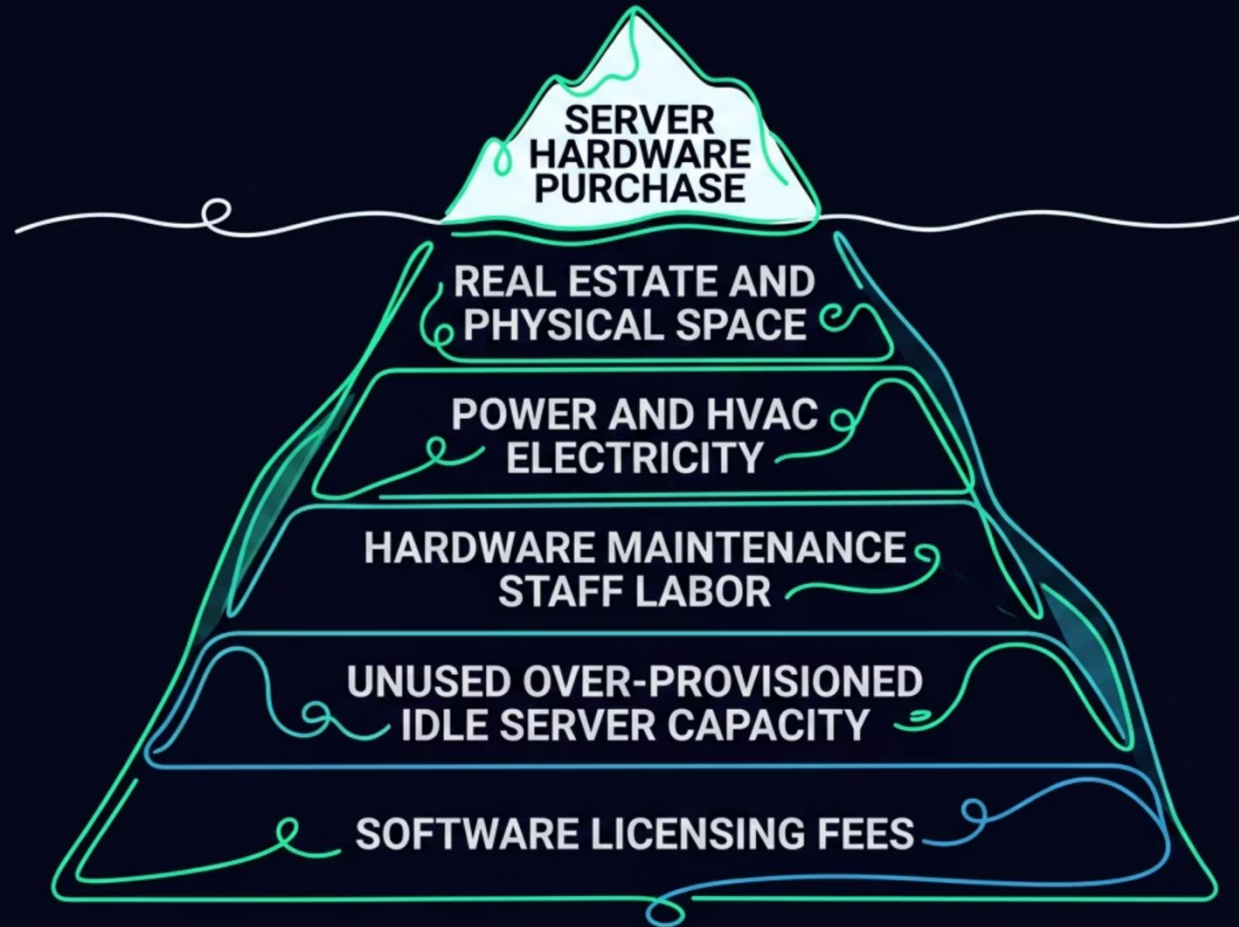
Total Cost of Ownership

Most On-Premises Costs Are Hidden Below the Surface

When a business owner says "I can buy a server for \$500 — why would I pay AWS monthly?" — they're only seeing the tiny tip of the iceberg. The real costs are underwater.



What's Hidden Below the Waterline?



TCO Breakdown

The Hidden Costs That Make On-Premises Expensive



Power & HVAC

Servers run 24/7. Industrial cooling prevents overheating. Both cost real money every month.



Idle Over-Provisioning

Businesses buy for peak capacity. That hardware sits unused and expensive 90% of the time.



Maintenance Staff

Someone drives to the office at 2 AM when a hard drive fails. That's labor — paid or unpaid.



Software Licensing

Operating systems, databases, and security tools carry ongoing licensing fees that add up fast.

TCO — The Cloud Advantage

How AWS Addresses Every Hidden Cost

AWS Owns the Building

Real estate, physical security guards, and facility management — all handled by AWS.

AWS Pays the Power Bill

Electricity and industrial cooling are AWS's problem, not yours. At massive scale, their efficiency is unbeatable.

AWS Replaces Hardware

Failed hard drives, failed networking cards — AWS data center staff handle all physical repairs around the clock.

You Pay for What You Use

No idle server overhead. Scale to zero at night, scale up for traffic spikes — pay only for actual usage.

Vocabulary Locked In

TCO, CapEx & OpEx — Defined

TCO (Total Cost of Ownership)

The *true* end-to-end cost of running a service — including all hidden indirect costs like power, cooling, staff, and real estate.

CapEx (Capital Expenditure)

Large upfront investment in physical assets. High risk — you're committed before you earn a dollar.

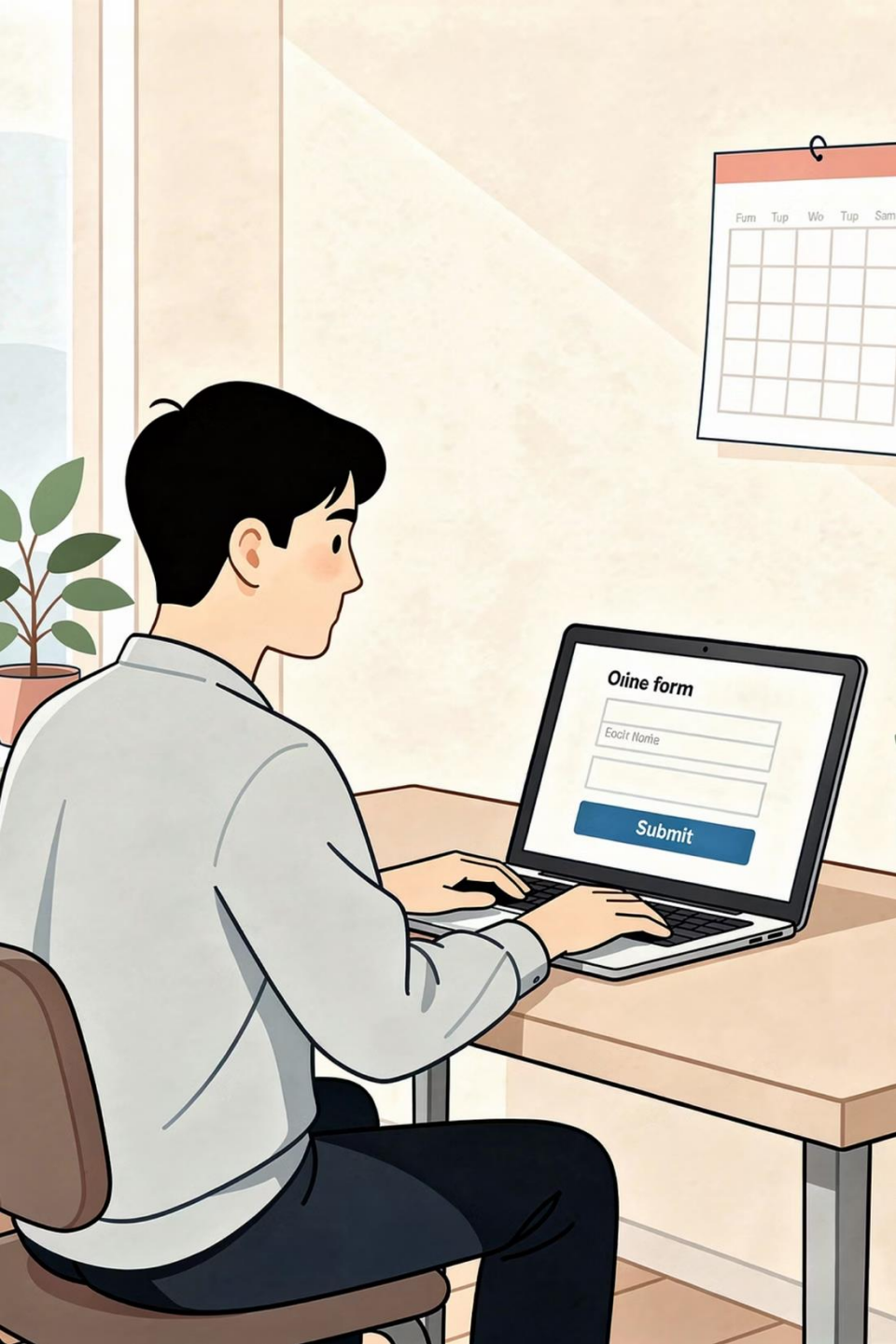
OpEx (Operating Expenditure)

Ongoing operational expenses billed by usage. Low risk — stop spending the moment you stop using.

SECTION 3

AWS Pricing Calculator

Estimate, configure, and present — before spending a single penny

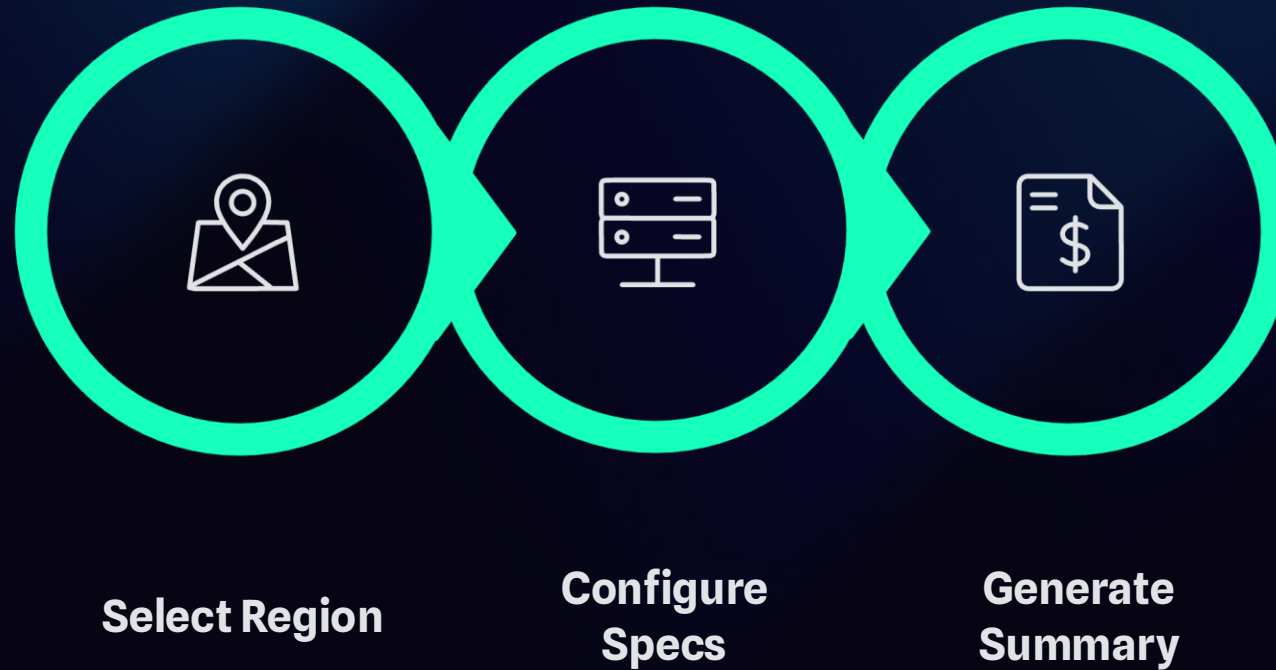


Your Professional Cost-Estimation Tool

The **AWS Pricing Calculator** is a free, web-based planning tool that lets you mock up entire cloud deployments before committing any budget. It's the exact tool your future employer will expect you to know.

AWS Pricing Calculator

Three Steps to a Professional Cost Estimate



The output is a clean, shareable web link you can send directly to your supervisor or client — audit-ready and professional.

Why This Tool Matters for Your Career

Show Up to Meetings With Numbers, Not Guesses

1 Select Your Region

Choose where your application will live — affecting both latency and price

2 Configure Compute & Storage

Set virtual CPU count, RAM, disk size, and data transfer volumes

3 Share the Estimate

Generate a permanent URL and send it directly to your manager or client



CASE STUDY

Real-World Problem Solving

Apply everything you've learned today to a real business scenario



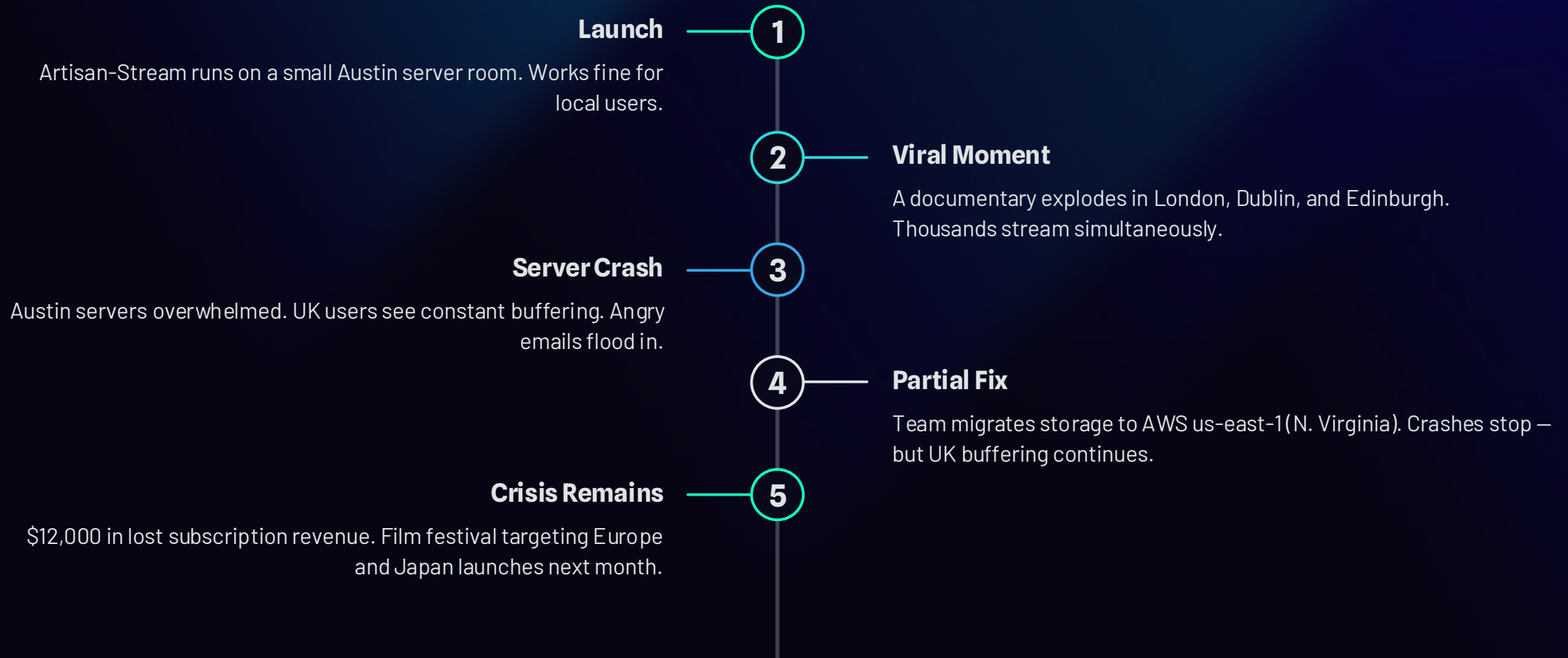
Vocational Case Study

The High-Cost Buffering Catastrophe

Artisan-Stream Inc. is a boutique video-on-demand startup streaming independent films. Their servers live in Austin, Texas. One weekend, a documentary went viral in London, Dublin, and Edinburgh – and everything broke.

Case Study — What Happened

Going Viral Exposed Every Infrastructure Weakness



Case Study — The Confusion

Why Didn't Moving to the Cloud Fix It?

Management's Assumption

"We moved to AWS — a premium cloud environment. Why are UK users still buffering?"

The Real Answer

us-east-1 is in Northern Virginia. Even on cloud infrastructure, the physical distance across the Atlantic Ocean introduces significant network latency. The cloud doesn't repeal the laws of physics. **Distance still matters.**



This is the most common misunderstanding among first-time cloud adopters — and why you are learning this today.

Case Study — Analysis Questions

Work Through These as a Team

1

Identify the Root Cause

Why did migrating to us-east-1 fail to solve buffering for viewers in the United Kingdom?

2

Propose an Architectural Solution

What specific AWS Global Infrastructure component should Artisan-Stream deploy for smooth playback in London and Tokyo — without duplicating full server infrastructure globally?

3

Evaluate the Financial Strategy

How does the proposed cloud solution change Artisan-Stream's financial risk compared to purchasing physical hardware in Europe and Asia?

Case Study — Solution Walkthrough

Question 1: The Root Cause



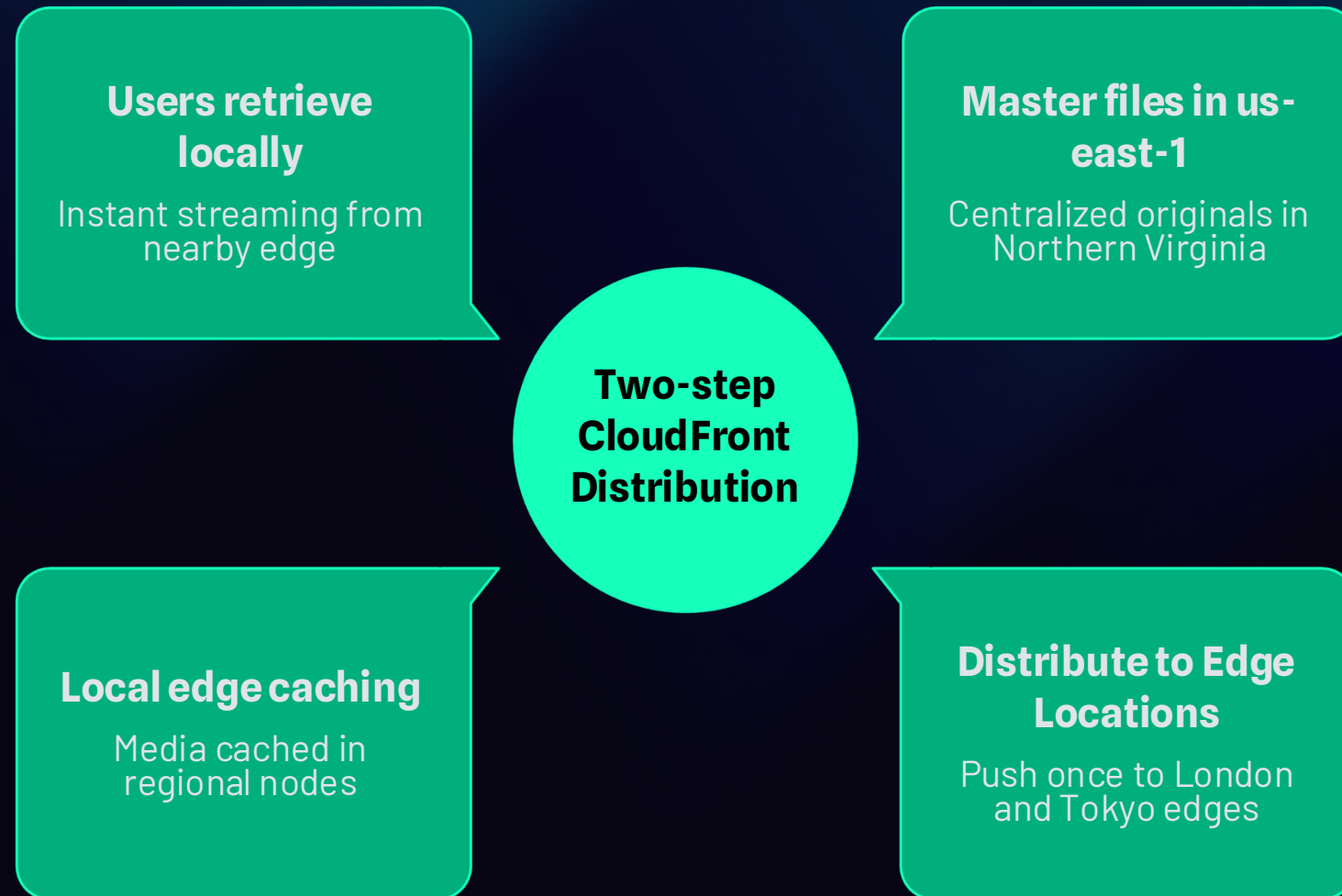
The Answer

us-east-1 is in Northern Virginia — thousands of miles from London. Even on cloud infrastructure, every mile of undersea fiber cable adds latency. The cloud server is faster and more reliable than the Austin server room, but it still has to send video data *across the Atlantic Ocean* for every UK viewer request.



Key insight: Cloud $\neq$ instant. Location still determines speed.

Question 2: The Right Architectural Fix



Deploy Amazon CloudFront with AWS Edge Locations. Media files are cached locally in European and Asian edge nodes. Users stream from their geographical neighborhood — not across an ocean.

Case Study — Solution Walkthrough

Question 3: The Financial Strategy Shift

Buying Physical Hardware in Europe & Asia (CapEx)

Requires massive upfront investment in servers, buildings, and staff across multiple countries. High permanent overhead — even on nights with zero film festival viewers. Drains venture capital reserves.

AWS CloudFront Edge Caching (OpEx)

Pay only for the **gigabytes transferred** during the film festival. When the event ends, costs drop to near zero. No leases. No idle hardware. No international staff. Pure utility billing.

Day 2 — Full Vocabulary Review

Every Term You Need to Own

Region

Distinct geographic area — fully isolated

Availability Zone

Independent data center(s) inside a Region

Edge Location / PoP

Local cache node for CloudFront CDN delivery

Latency

Delay caused by physical distance traveled

Data Sovereignty

Legal requirement keeping data within borders

TCO

True end-to-end cost including all hidden costs

CapEx

Upfront investment in physical assets

OpEx

Ongoing pay-as-you-go operational expenses

You've Got This — Key Takeaways for Today

01

Infrastructure is physical

Regions, AZs, and Edge Locations are real buildings — location determines latency and legal compliance.

02

Cloud economics favor business

Replacing CapEx with OpEx eliminates hidden TCO costs and aligns spending with actual usage.

03

You can estimate before you spend

The AWS Pricing Calculator turns infrastructure planning into a professional, shareable deliverable.

❏ Next up: 15-minute break, then the AWS Pricing Calculator hands-on lab. Come back ready to build.

